

m6A perturbation in mouse bone marrow — preliminary single-cell figures

Charles project — internal, exploratory

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Preliminary analyses — internal discussion only. Mouse bone-marrow scRNA-seq, ~42,000 cells, 2 × 2 design: genotype (WT, Mutant) × treatment (DMSO vehicle, STM2457 METTL3 inhibitor), two biological donors (paired blocking factor; WT and Mutant separated *in silico*). Cell types use the canonical `cell_type` annotation (HemaScribe lineage calls + manual marker validation, 32 types). Composition was modelled with scCODA, differential expression with pseudobulk PyDESeq2 (~ donor + genotype + treatment), and pathway activity with decoupler ULM on the DESeq2 statistic against MSigDB Hallmark/Reactome/KEGG. The genotype contrast is Mutant relative to WT; the treatment contrast is STM relative to DMSO. Figures are descriptive; see caveats at the end.

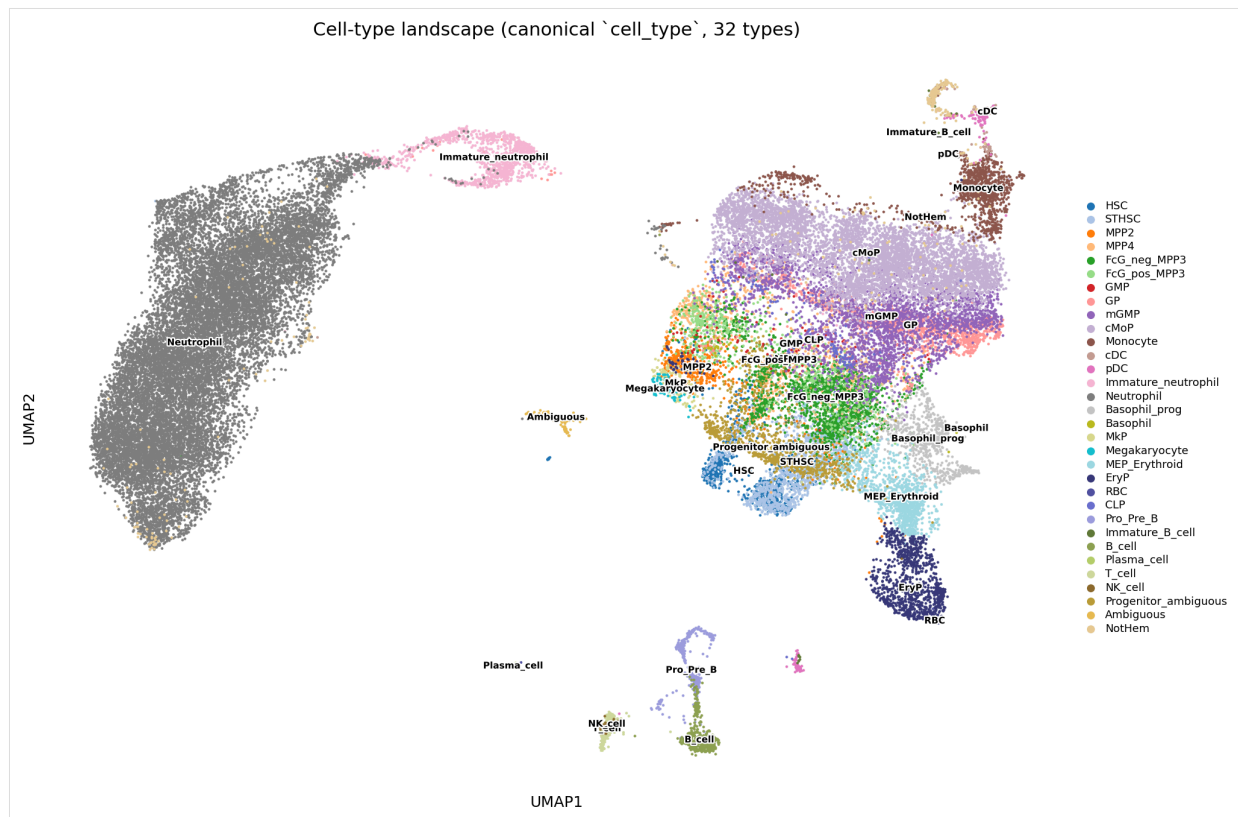


Figure 1. Cell-type landscape. UMAP of ~42,000 bone-marrow cells coloured by the canonical `cell_type` annotation (32 types). Neutrophils are the dominant population (~42%); the stem/progenitor

compartment (HSC, STHSC, MPP, GMP) and the lymphoid and erythroid compartments are resolved separately.

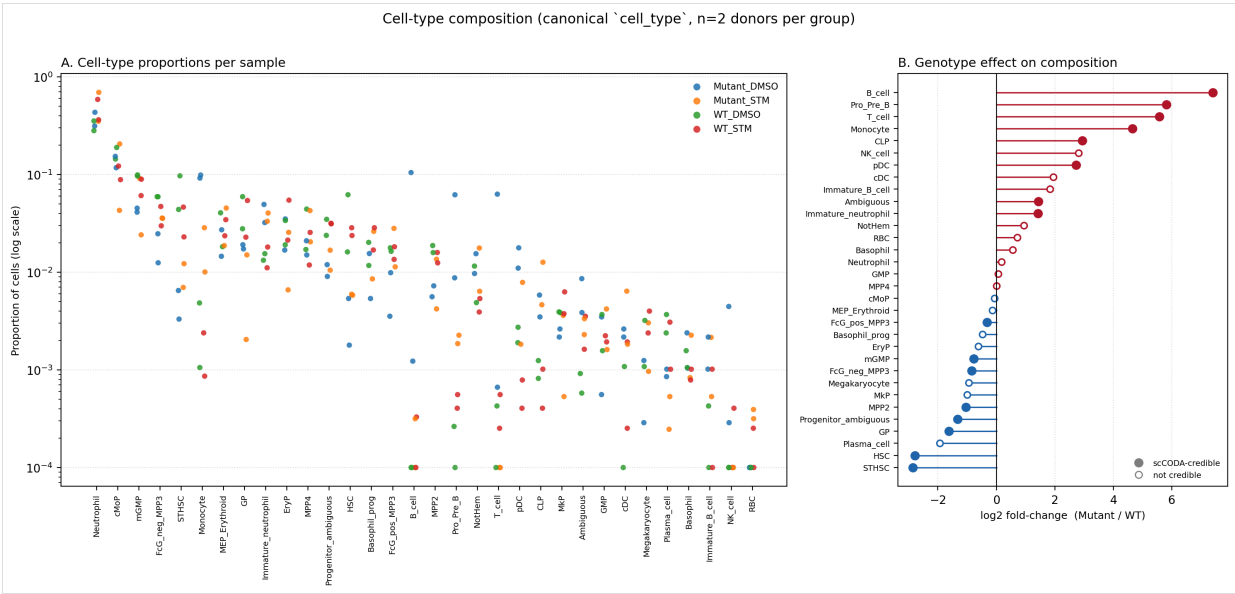


Figure 2. Cell-type composition. (A) Per-sample proportion of each cell type on a log scale, coloured by condition (two donors per condition). (B) Mutant-vs-WT log2 fold-change per cell type, pooled over treatment; filled points are flagged credible by scCODA in the genotype contrast, hollow points are not. The stem/progenitor populations HSC and STHSC are lower in Mutant; Monocyte is higher in Mutant. The largest lymphoid fold-changes (B cell, T cell, Pro/Pre-B) coincide with populations captured in only a subset of samples and partly reflect donor/capture differences rather than genotype.

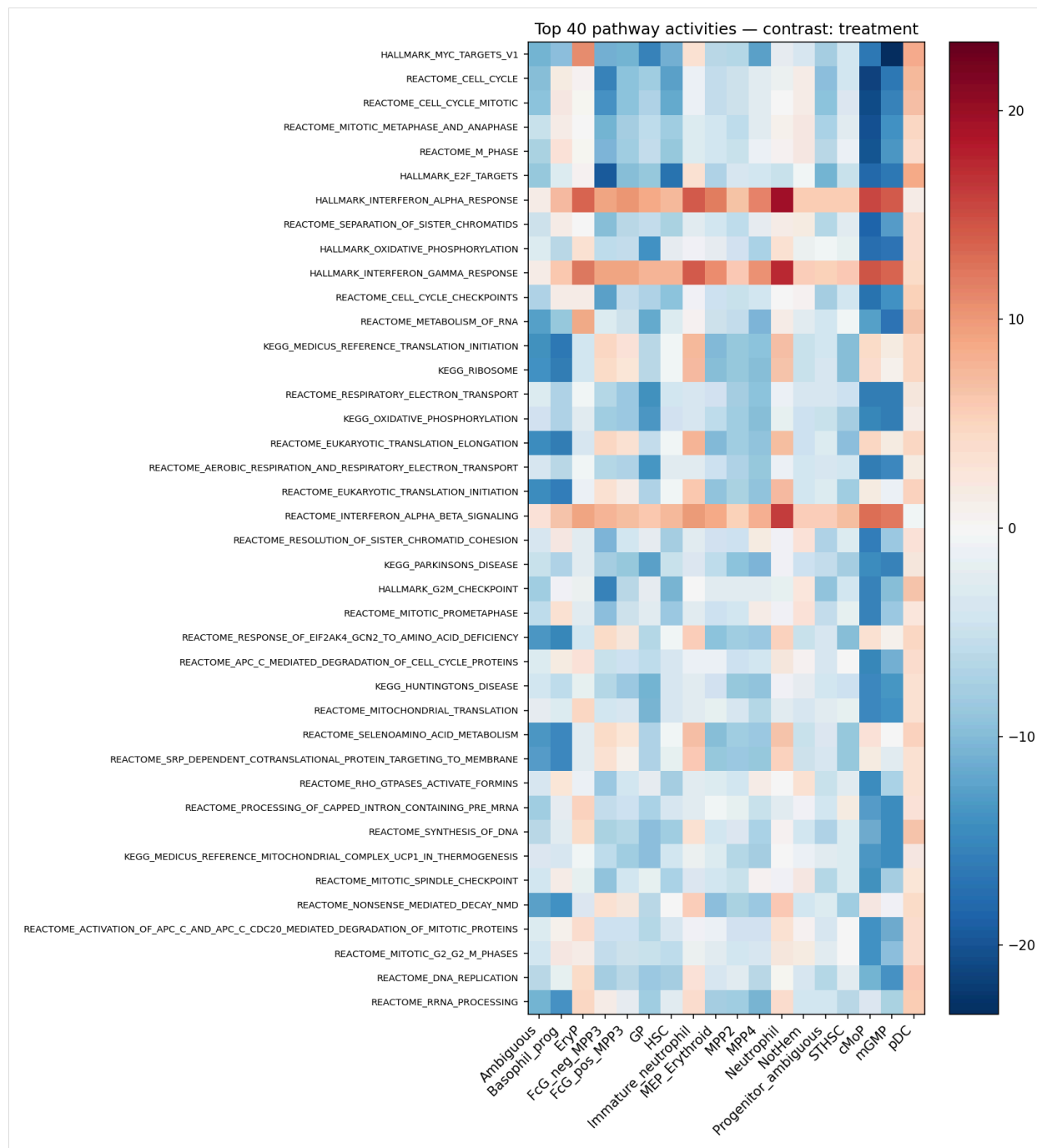


Figure 3. Pathway activity, STM versus DMSO. Top-40 pathway activities (cell types × pathways) from decoupler ULM. Negative scores indicate lower activity under STM. Translation, ribosome, and rRNA-processing programmes, together with MYC/E2F proliferation programmes, are coordinately lower under STM across progenitor populations.

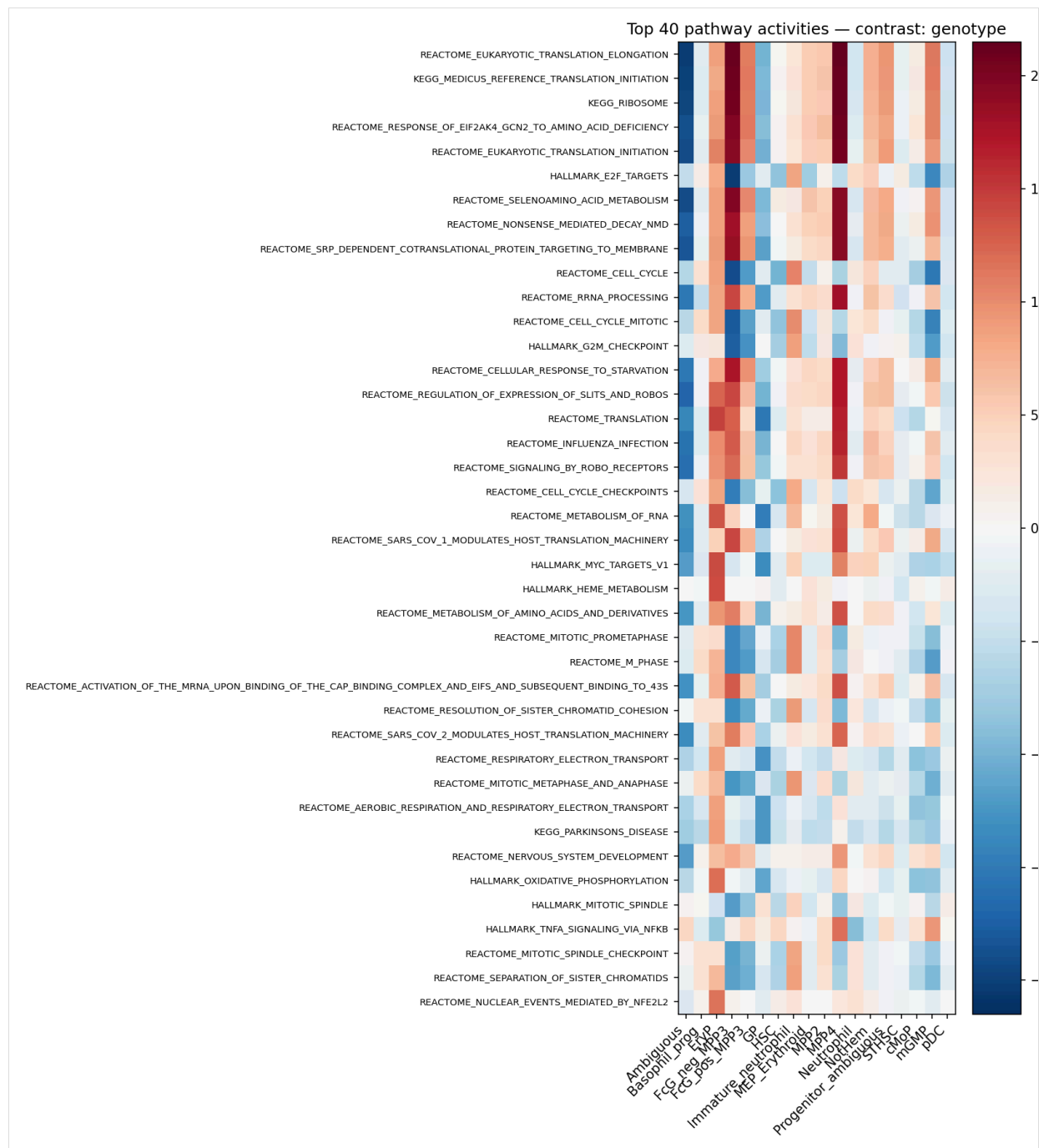


Figure 4. Pathway activity, Mutant versus WT. Top-40 pathway activities. Negative scores indicate lower activity in Mutant. In HSCs the type-I interferon response is lower in Mutant; the cell-cycle/E2F/G2M direction in HSC is sensitive to how HSC is defined (it inverts between the restrictive HemaScribe and broader manual gates) and is shown for completeness only.

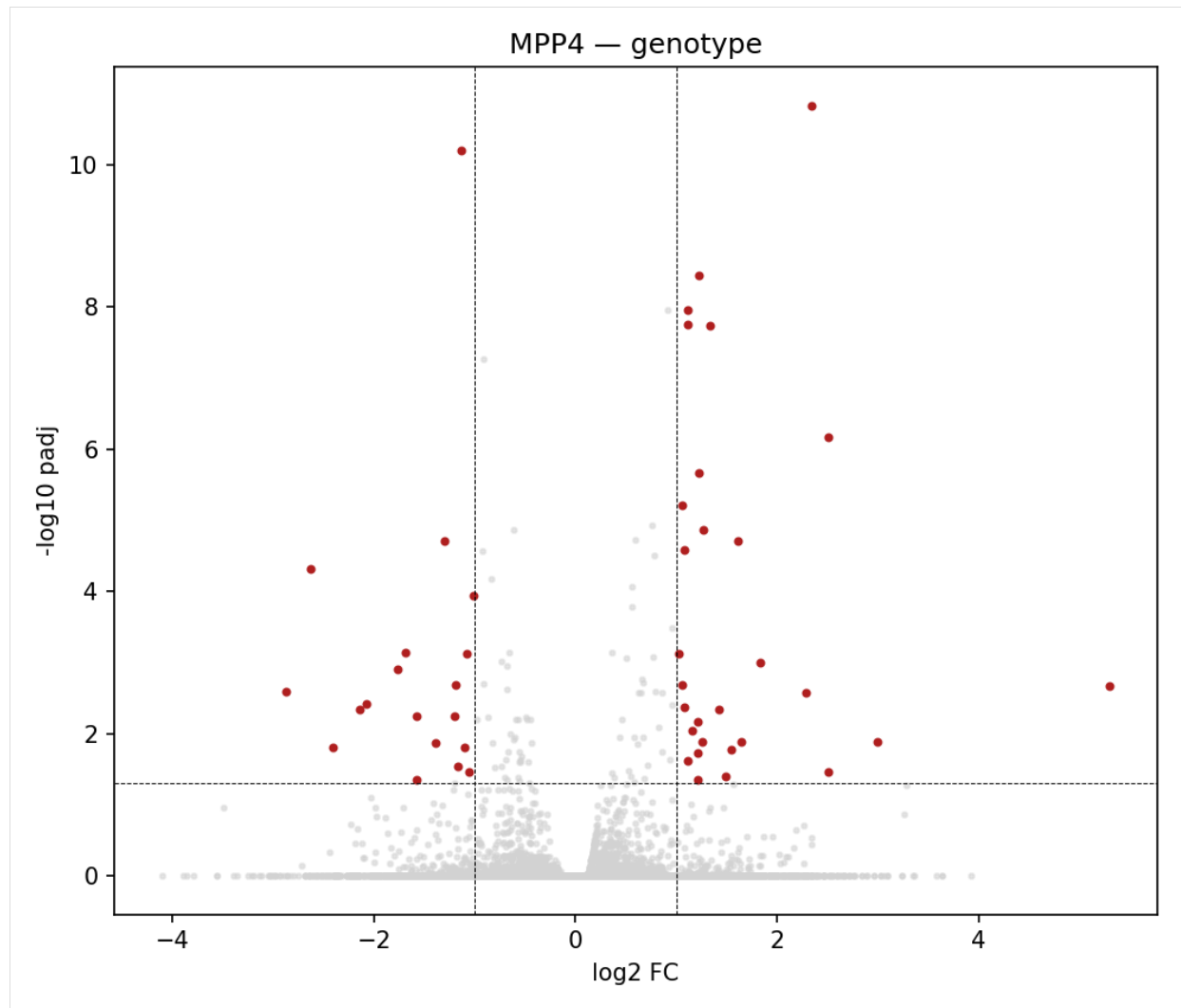


Figure 5. Differential expression, MPP4 — Mutant versus WT. Pseudobulk PyDESeq2 volcano (positive log2FC = higher in Mutant). Genes higher in Mutant include *Irf8*, *Ccl4*, *Il12a*, and *Dntt*; this was the largest genotype DEG response (48 genes at padj < 0.05, |log2FC| > 1).

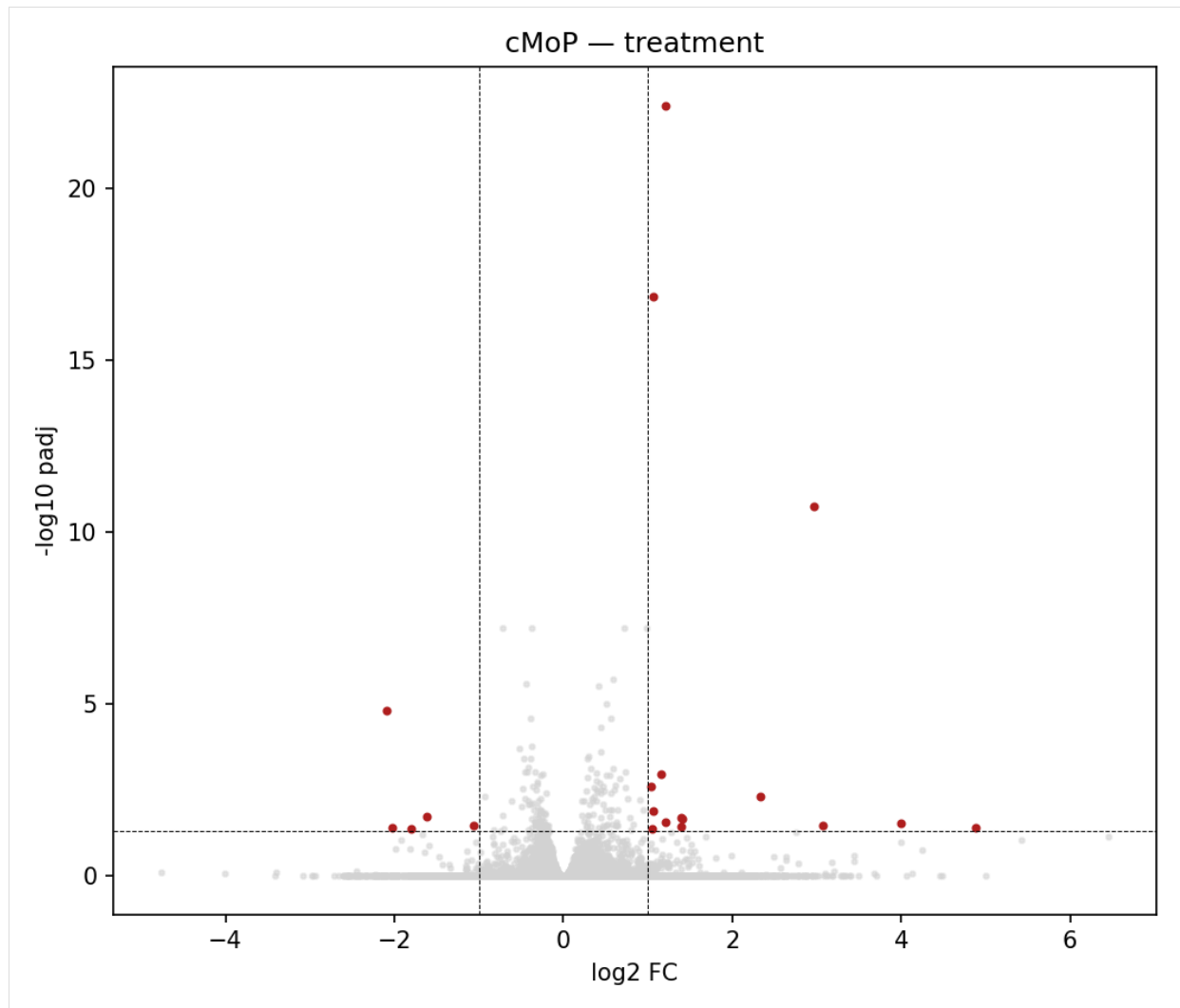


Figure 6. Differential expression, cMoP — STM versus DMSO. Pseudobulk PyDESeq2 volcano (positive log2FC = higher under STM). Genes higher under STM include *Egr1*, *Gbp2*, *Cd74*, and *Mcpt8*.

Caveats

- These are exploratory, hypothesis-generating analyses; no claim is confirmatory.
- Two biological donors only — low power, especially for the treatment contrast; no genotype × treatment interaction was modelled.
- Genotypes were separated *in silico*, not by physical sorting, so the genotype axis depends on assignment accuracy.
- Sorting and capture differences confound some populations (notably the lymphoid compartment) and inflate apparent donor effects.
- A few clusters retain residual impurity, so cell-type-resolved claims for the rarest populations (basophils, megakaryocytes) rest on small pseudobulk counts.
- scCODA credibility is directional, not a calibrated effect size.

Reproducible from results/18_composition/final_annotation,
results/19_pseudobulk_deg/final_annotation, and
results/20_pathway_activity/final_annotation. Code:
pipeline/scripts/18_composition_sccoda.py, 19_pseudobulk_deg.py, 20_pathway_activity.py;
figures report/make_fig1_umap.py, report/make_fig2_composition.py. Canonical annotation:
21z_finalize_annotation.py, 22_clean_annotation_columns.py. Branch feat/composition-
pseudobulk.